1. INTRODUCTION

Social Media platforms such as Twitter, Facebook, Instagram, etc. have become an important part of our everyday lives. They help in connecting people across the world and provide a platform for communication. People have the freedom to express their opinions on a larger scale through these platforms.This is an advantage of social media, but can also mislead people.

Covid-19, caused by the novel Coronavirus, was declared as a global pandemic by the World Health Organisation on the 11th of March 2020. It has had a great impact on all our lives. The Social media platforms have played a significant role in helping us out during these tough times. But, they have also been the platform for spreading false information. For example, statements like-
"If you take crocin thrice a day you are safe from coronavirus", are considered fake news, which are used to mislead the readers.

In this report, we describe our approaches evaluated for Constraint@AAAI2021 - COVID 19 Fake News Detection in English.Given a social media post, the objective of the task is to classify it into either fake or real news. The data set was collected from various social-media platforms such as Twitter, Facebook, Instagram, etc. It is a text based classification task.

We began with Machine Learning models for the classification with tf-idf vectors of the data as their input. This was followed by usage of Word2Vec for vectorisation and recurrent neural networks as our deep learning model. The other technique we used was a pre-trained model based on BERT.

1.1 Problem Statement:

Detection of COVID-19 related fake news in English. Given a social media post, the objective of the shared task is to classify it into either fake or real news. For example , the following two posts belong to fake and real categories, respectively.

If you take Crocin thrice a day you are safe . – **Fake**

Wearing a mask can protect you from the virus. – **Real**

2. LITERATURE SURVEY

The problem of fake news detection has been prevalent for quite some time. It has been studied by several researchers. One of the ways can be finding a user's credibility by analyzing their followers, the number of followers, and their behavior as well as their registration details. In addition to these details [6] have used other factors such as attached URLs, social media post propagation features, and content-based hybrid model for classifying news as fake or genuine.

Besides using the characteristics of user-patterns and user details who share fake news, and the number of followers they have in the social media platform, another context useful for classifying any social media news post is the comments section. [5] have performed a linguistic study and have found comments like "Really?", "Is it true?" in the comment section of some of the fake posts. They have further implemented a system that clusters such enquiry phrases in addition to clustering simple phrases for classifying rumors. But the dataset on which we are performing classification has only the tweets, hence their methodology can not be applied on our data set.

Apart from user-context and social-context features the content of the fake news has also been a proven way of detecting fake news and rumors. [2,3] provide methods to classify the tweets by using only the contextual data from the tweets. Data cleaning is the first followed[2,3]. It includes removal of punctuation, stop words, special characters, etc. After cleaning we can not feed this cleaned text directly into the algorithms. Most algorithms expect numerical feature vectors, and the cleaned text is of the form string [2]. In the paper "The Detection of Fake Messages using Machine Learning " [2], the function TfidfVectorizer is used for this purpose. The database used here consists of tweets posted around the election of September 12, 2012 and they have proposed a method to classify those tweets as true or false. Thus the vectorized form of input data is then fed into the Classifiers such as Linear Support Vector Machines (LSVM), Naïve Bayes (NB), Decision Trees (DT), ExtraTrees (ET), Stochastic Gradient Descent (SGD) and Random Forests (RF). Each algorithm has been trained and validated using the 10-fold cross-validation method. And the Decision Tree algorithm seems to perform the best for this classification with an F-score of 88%. We used the same methodology on our data set to perform the classification of tweets related to Covid-19.

3. METHODOLOGY

3.1 Support Vector Machine (SVM)

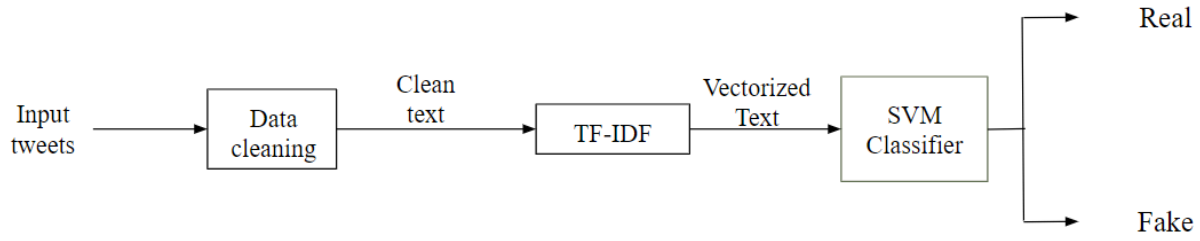

Fig1. Block diagram of SVM

3.1.1 Data Cleaning:

Text preprocessing is important because it transforms the text into more digestible form, so that algorithm can perform better. Following are the preprocessing techniques that we performed on our dataset:

3.1.1.a Removing of the HTML tags and urls:

Often in the process of gathering a dataset, web or screen scraping leads to the inclusion of HTML tags in the text. These tags are often not paid heed to but it is necessary to get rid of them.

3.1.1.b Lowercasing:

The idea of lowercasing is to convert the input text into the same casing format so that ‘text’,’Text’ and ‘TEXT’ are treated the same way.

3.1.1.c Removal of Punctuations,Special characters:

One another common text preprocessing technique is to remove the punctuations from the text data. This is again a text standardization process that will help to treat 'hurray' and 'hurray!' in the same way. : Special characters are not readable because they are neither alphabets nor numbers. They include characters like ”*”, ”&”, ”$”, etc.

3.1.1.d Stopwords Removal:

Stopwords are very common words. Words like “we” and “are” probably do not help in all NLP tasks such as sentiment analysis or text classification. Hence, we can remove stopwords to save computing time and efforts in processing large volumes of the texts.

3.1.1.e Lemmatization:

Lemmatization is the process of converting text to its base form. For example, “caring” to “care”.

0	The CDC currently reports 99031 deaths. In general the discrepancies in death counts between different sources are small and explicable. The death toll stands at roughly 100000 people today.
1	States reported 1121 deaths a small rise from last Tuesday. Southern states reported 640 of those deaths. https://t.co/YASGRTT4ux
2	Politically Correct Woman (Almost) Uses Pandemic as Excuse Not to Reuse Plastic Bag https://t.co/thF8GuNFPe #coronavirus #nashville
3	#IndiaFightsCorona: We have 1524 #COVID testing laboratories in India and as on 25th August 2020 36827520 tests have been done : @ProfBhargava DG @ICMRDELHI #StaySafe #IndiaWillWin https://t.co/Yh3ZxknnhZ

Table 1(a): Initial data

0	cdc currently report death general discrepancy in death count between different sources small explicable death toll stand rough people today
1	report death small rise from last tuesday southern report death
2	politically correct woman almost use pandemic excuse not reuse plastic bag coronavirus nashville
3	indiafightscorona test laboratories india august test done indiawillwin

Table 1(b): Cleaned data

The cleaned text is then vectorized using ‘term frequency-inverse document frequency’, shortened TF-IDF.
TF-IDF, which stands for term frequency—inverse document frequency, is a scoring measure. TF-IDF is intended to reflect how relevant a term is in a given document.
Term Frequency, which measures how frequently a term occurs in a document. TF(t) = (Number of times term t appears in a document) / (Total number of terms in the document).

Inverse Document Frequency, which measures how important a term is.
IDF(t) = log_e(Total number of documents / Number of documents with term t in it).

The tf-idf value for a word t in tweet d is:

TF-IDF(t,d) = tf(t,d) x idf(t,d)

This calculated TF-IDF value gives us a way to associate each word in a tweet with a number that represents how relevant each word is in that tweet. Then, tweets with similar, relevant words will have similar vectors, which is what we are looking for in a machine learning algorithm. The highest scoring words of a tweet are the most relevant to that tweet, and therefore

they can be considered keywords. The output of tf-idf can be observed in Table 2.

(A, B)	C
(0,1364)	0.18981
(0,2052)	0.24457
(0,2200)	0.21829
(0,2286)	0.23843

Table 2. A: ID index (tweet), B: Feature index, C: tf -idf value

This vectorised data is then fed into a support vector machine classifier with a linear kernel. And the parameters were set using grid search CV. The objective of the support vector machine is to find a hyperplane in an N-dimensional (N- number of features) space that distinctly classifies the data points. Having as many features (thousands of words and each unique word becomes a feature) using a non-linear will overfit the data hence we use a linear kernel. SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence the algorithm is termed as Support Vector Machine. Here, SVM will generate hyperplanes iteratively that segregates the 2 classes in the best way.Then, it will choose the hyperplane that separates the classes correctly.

3.2 Word2Vec with LSTM model

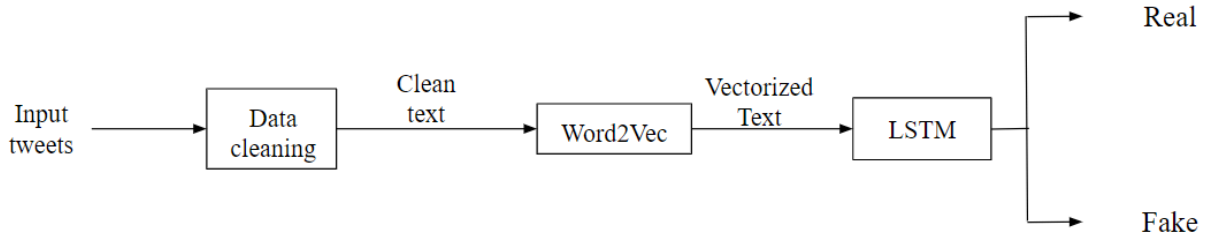

Fig3. Block diagram of LSTM based classifier

Word2Vec is the word embedding technique. It is the vector space representation of the words that provide projection where words with similar meanings are clustered within the space. LSTM is Gated-RNN architecture along with the feedback connections. LSTM is more efficient

to capture the words order and semantic relationships between them, which is usually critical to a text's meaning.We have used same data cleaning method as done in 3.1.1.Then we have passed the cleaned text to the word2vec where it trains the model for data which is corpus(tokenized tweets). Semantic vectors provided by Word2vec preserve most of the relevant information about a tweet while having relatively low dimensionality which allows better learning. Then the output is fed to the LSTM model for learning which is followed by two dense layers ending with the sigmoid classifier. The limitation of the model is, it did not take the context of the word into account. Because the same word may have different meaning in different contexts, but Word2Vec gives the same meaning for the words in both the contexts.

3.3 BERT base

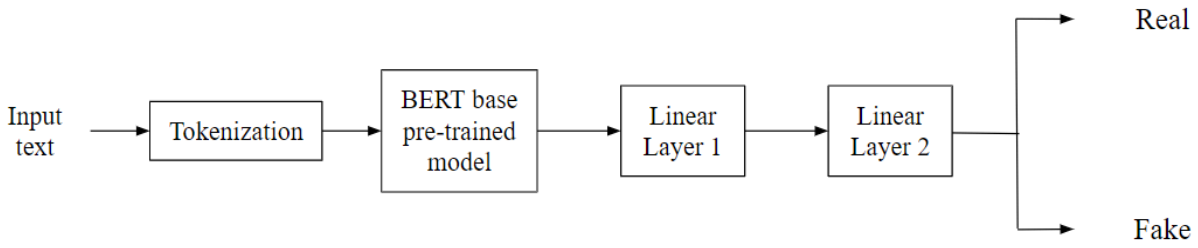

Fig3. Block diagram of BERT-base model

BERT-base [4] is a model that contains 12 transformer blocks, 12 self-attention heads, and a hidden size of 786.It is trained on lower-case english words.The major component in the transformer is self attention which is used for generating a contextual embedding of any given word in the input sentence with respect to the other words. The input for BERT contains embeddings for a maximum of 512 words and it outputs a representation for this sequence. We used BERT to extract the embedding from each tweet in the dataset and then used these embeddings to train the text classification model.Since BERT makes use of transformer, Transformer encoder reads the entire input text or words sequence at once which allows the model to learn the context of a word based on all of its surroundings. We tokenized the input tweet and fed it to a pre-trained BERT-base model followed by two linear layers with dropout and ReLU activations.

4. EXPERIMENTS

4.1 Data Description

Covid-19 fake news detection dataset from the challenge given by Codalab consists of tweets and their corresponding label. The label categorizes the tweets as either fake or true.The dataset has a predefined train, test, and validation split. The train data has 6420 samples, test data has 2140 samples and validation data has 2140 samples; making it a total of 10,700 media articles and posts acquired from multiple platforms.Train data contains 3060 fake samples and 3360 real samples while validation and test data contains 1020 fake samples and 1120 real samples each.There are no missing values in the datasets. Every tweet in the train and validation dataset is unique, i.e. there is no redundancy in the tweets.

Fig4(a). Train data

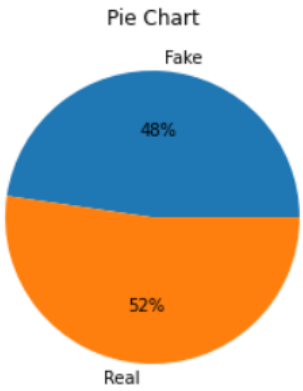

Fig4(b). Validation data

4.2 EDA

We analysed the top 20 frequently used Hashtags in the tweets, from Fig 5(a), we can see that hashtags like #coronavirus, #indiafightscorona, #staysafe, etc. occur most frequently. We also analysed the most frequently occuring words which can be seen in 5(b), which says that words like covid, cases, coronavirus, etc. occur most frequently.

Then the most frequently tagged id's were also studied and the Fig 5(c) shows that id's like @MoHFW_INDIA, @ICMRDELHI, etc. are the most tagged ones.

We also explored the frequently used hashtags in Fake and Real tweets separately. From Fig 5(d,e), we can see that hashtags like #coronavirus, #indiafightscorona, #coronavirusfacts are common in both fake and real tweets and hence do not hold much significance while performing the classification task.

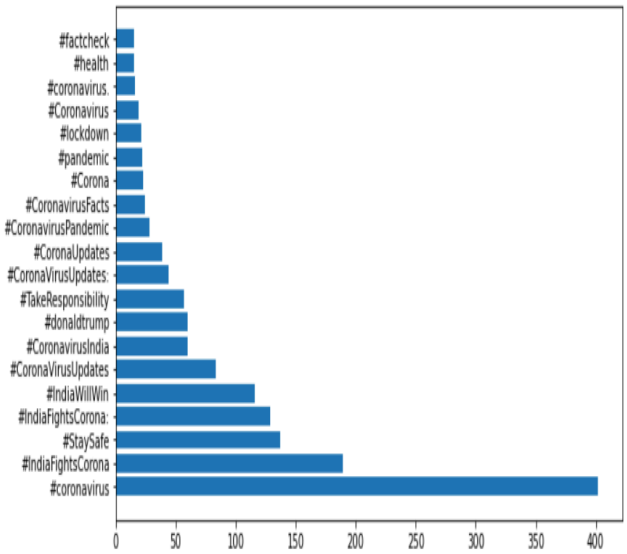

Fig 5(a) Hashtags

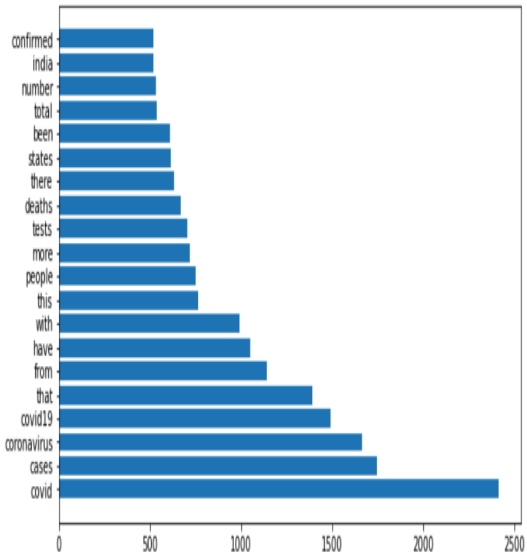

Fig 5(b) Common Words

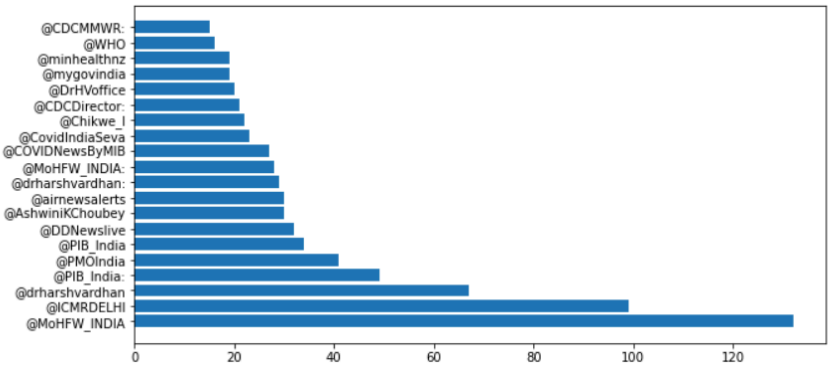

Fig 5(c) Tagged ID's

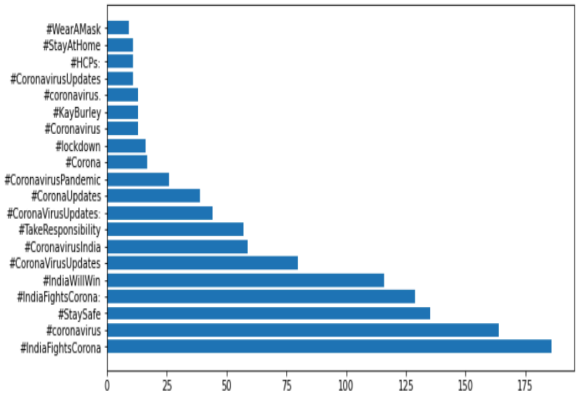

Fig 5(d) Hashtags from Fake tweets

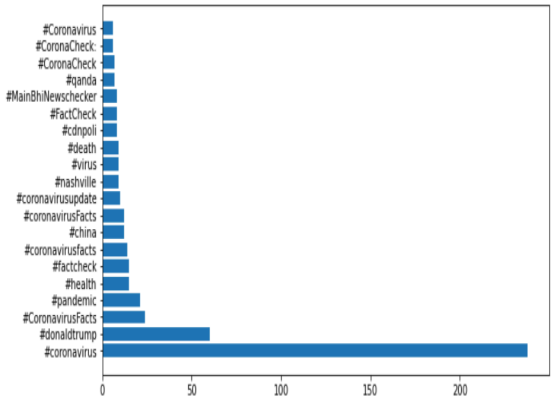

Fig 5(e)Hashtags from Real tweets

4.3 Implementation Details

We have set the maximum length for tokenization as 128. The batch size used is 32. We have trained the model for 10 epochs with the learning rate 0.0002 and Adam optimizer. Loss is calculated using cross entropy loss.

4.4 Results and Discussion

	Classifier	F1 score
1	ML based Classifier	0.91
2	Word2Vec + LSTM based Classifier	0.92
3	BERT-base	0.98

Table 3: F1 scores of all the classifiers

Our evaluation metric is F1 score. Classification performance of all the models is shown in table 3 which shows that the BERT-base model performs the best with an f1 score of 0.98. The Machine learning based model gives us an f1 score of 0.91 and 0.92 f1 score was obtained by the LSTM based classifier. We can notice that the pre-trained model performs better when compared to the other models since it is trained on larger data and helps quicken the process of training as well.

Fig 5(a), that is the bias-variance tradeoff for the machine learning classifier shows that the model has low bias and low variance. In Fig 5(b) as well we notice similar bias-variance tradeoff for the LSTM based model. In 5(c), that depicts the bias-variance trade off for the BERT-base

model we notice that it has low bias and constant low variance.

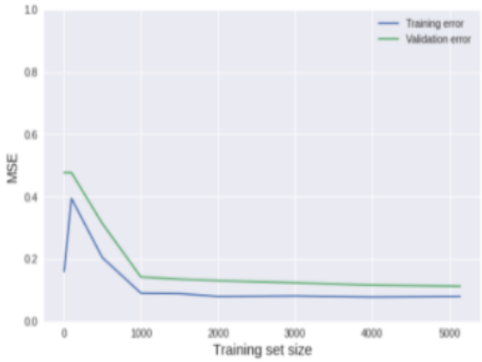

Fig 5(a) . Bias variance tradeoff for SVM classifier

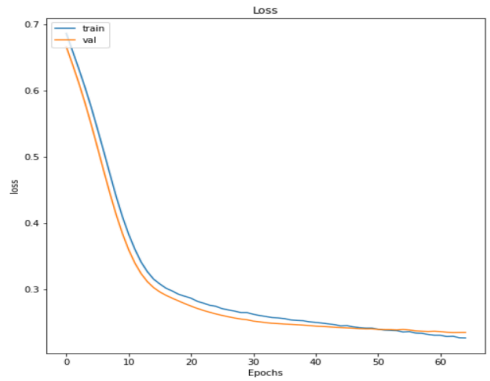

Fig 5(b).Bias variance tradeoff for Word2vec Model with LSTM

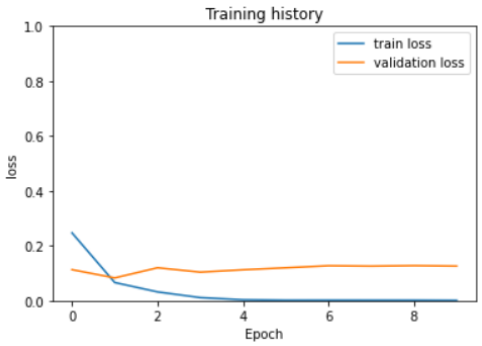

Fig 5(c).Bias variance tradeoff for BERT-base Classifier

5.CONCLUSION

In this report under the shared task of Contraint@AAAI 2021 Covid-19 Fake news detection, we describe methodologies to classify tweets related to Covid-19 as fake or real. The data set made available by the Codalab platform consisted of labelled tweets related to Covid-19. We experimented with three methods to perform the classification, one with tf-idf vectors fed into the SVM model which gave us an f1 score of 0.91. Two, Word2Vec for vectorisation and an LSTM based network for classification which gave an f1 score of 0.92. The best results were obtained with the pre-trained model of BERT-base with an f1 score of 0.98.

Here our data set was limited to tweets, but while performing the classification some more user account details like number of followers, whether the account is active or not, etc. could improve the training process and help the models perform better.

6.REFERENCES:

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